

# Jalay Rupera

Bangalore, Karnataka · jalayrupera13@gmail.com · +91 82381 45144  
github.com/jalayrupera · linkedin.com/in/jalay-rupera-b774b5131

## SUMMARY

Backend engineer with **4.5+ years** building scalable, distributed Python systems and AI/LLM-powered services. Founder-level ownership spanning REST API design, asynchronous pipelines (Celery, Redis, RabbitMQ), and container orchestration (Docker, Kubernetes). Shipped production systems serving **10K+ concurrent users**, processing **40K+ patient eligibility checks monthly**, and contributing to **\$1.5M+ in ARR**.

## CORE SKILLS

- **Languages:** Python, TypeScript, SQL
- **Backend:** FastAPI, REST API design, microservices, system design
- **Data & Messaging:** MongoDB, Redis, Celery, RabbitMQ
- **AI / LLM:** LLM prompting, structured data extraction, AI workflow integration
- **Infrastructure:** Docker, Kubernetes (K3s), AWS, Azure, CI/CD, observability
- **Tools:** Playwright, Git, Linux, ROS2, MQTT

## EXPERIENCE

### Software Development Engineer, Founding Team — Mantys

Apr 2025 – Present · Bangalore, KA

- Build automated web-scraping pipelines in Python and Playwright that **cut manual patient-data retrieval time by 80%** across hospital workflows.
- Design LLM-based transformation workflows that convert unstructured HTML into structured JSON at **95%+ accuracy**, powering **40K+ real-time patient insurance eligibility checks monthly**.
- Ship features and bug fixes to the core Chrome extension, **reducing user-reported issues by 90%**.
- Partner with cross-functional teams to strengthen backend reliability, observability, and system performance.

### Founder & CEO / Lead Backend Engineer — Vigilare Technologies

Jan 2024 – Feb 2025 · Bhavnagar, GJ

- Founded and led a defense-tech startup building a GPS-denied autonomous drone system for mapping and inspection.
- Architected backend services and REST APIs in Python and FastAPI for real-time telemetry streaming and mission control.
- Built ROS2 packages for sensor fusion, Visual-Inertial Odometry (VIO), and MQTT-based telemetry.
- Containerized and deployed the platform with Docker and K3s, enabling modular, scalable releases.

### Backend Developer, B2B SaaS — Zu Technologies

May 2022 – Sep 2023 · Bangalore, KA

- Developed production-grade REST APIs and microservices with Python (FastAPI) and MongoDB.
- Led the backend for "Stock Wars," a virtual stock-trading game **serving 10K+ concurrent users**, using Redis caching and message queues to sustain performance.
- Integrated Celery, Redis, and RabbitMQ for asynchronous job scheduling and background task execution.
- Deployed services on Docker and Kubernetes, **improving reliability and deployment speed by 40%**.

### Software Developer — Mindtree Limited

Jul 2021 – Feb 2022 · Bangalore, KA

- Built backend APIs and database schemas in Python for internal enterprise applications.
- Drove performance optimization and backend automation across client projects.

## PROJECTS

### Insurance Eligibility Product — Mantys

- Shipped (with the CTO) a production Chrome extension used by UAE hospitals to verify insurance eligibility in real time via Emirates ID, Member ID, or DHA ID, integrating with hospital information systems (HIS) to automate form filling and document uploads.
- **Cut front-desk workload and paperwork errors by 90%**, contributing to **\$1.5M+ in company ARR**.

### Drishti — Autonomous Drone System — Vigilare Technologies

- Built a GPS-denied navigation stack (sensor drivers, VIO, backend) with Python/FastAPI telemetry pipelines, deployed via Docker and K3s.

### Stock Wars — Virtual Trading Platform — Zu Technologies

- Designed a real-time trading-simulation backend with FastAPI, Celery, Redis, and RabbitMQ, optimized for latency-critical APIs.

## EDUCATION

B.Tech, Computer Science — REVA University, Bangalore · 2017 – 2021